

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

In this lab we began learning the basics of Vivado. Following the procedure we started a new project, loaded our first file, edited it, and ran it in simulation as well as loaded it onto the basys3 board. We were able to successfully light up an LED that is controlled by a switch on the board.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

- Create project file
- Write code
- Run simulation
- Generate bitstream/upload to device
- See it working in real life

2 - What is the value in looking at the elaborated design schematic?

- It allows us to visualize the circuit that we created with code. This is important in the learning process as it gives a different perspective.

3 - Why should we simulate our designs frequently? What does the simulation do?

If your code is simulated regularly, you keep your code in smaller segments, which makes it easier to troubleshoot. The simulation attempts to run the program in its current state, and will tell you what errors or problems arise with your program, which allows you to go back and fix it.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.